Ghoti Labs - Factory ATE Bringup Notes

GHT-BU-SPI-014, rev 3

g.hoti, 2026-02-14

Winbond W25N01GV (128 MiB usable). Same part we used on the B6 boards so most of this is copy paste from AKR-BU-SPI-010 -- if something looks off blame me.

page size 2048B

oob 64B per page

erase block 128KiB (64 pages)

ecc: on-die BCH8, 512B sector

128 MiB / 128 KiB = 1024 PEBs

Region map for the shipping image (post-strip offsets, don't forget the raw dump has OOB every page):

```
0x00000000   256K   SPL
0x00040000   768K   U-Boot main
0x00100000   128K   env0
0x00120000   128K   env1  (backup, redundant format)
0x00200000   3M     ubi.rootfs
0x00500000   ~11M   erase state, unallocated
0x00FF0000   32B    OTP-mirror   <-- see below
0x01000000          EOF (16MiB logical window)
```

OTP-mirror:

This one moved between rev 2 and rev 3. Used to live at 0x00F80000 but we had that DVT-12 unit fail bringup because the bootROM was locking two PEBs there for its own use (still not sure why, need to ask Karp). New location is 0x00FF0000. Size unchanged at 32 bytes.

Fields:

```
region:  OTP-mirror
offset:  0x00FF0000    (was 0x00F80000 pre-rev2)
size:    32
access:  /dev/mtdblock0 (raw, bytes 16711680..16711711)
ro:      yes -- bootROM enforces after PEB0 lock
```

Written by ATE fixture GHT-ATE-04 at wafer test. Content is per-device (SHA of fuse array + wafer/lot id, silicon team owns the derivation). Do not relocate again without re-signing the diagnostic ramdisk -- derive.sh has the offset hardcoded I think? Karp said no, it reads from this doc, but I'm not sure. Verify before rev 4.

FIT image:

```
/boot/image.fit inside the ubi.rootfs volume
conf@1 = production. signed (sha256+rsa2048, key hint 'ghoti-prod').
         boots ramdisk@1.
conf@2 = diagnostic. NO signature node. boots ramdisk@2.
```

conf@2 should have been dropped before the B7 mask ROM cutover -- see change request GHT-CR-088. As of this rev it's still in there. Manufacturing knows. If you're reading this because you're doing post-mortem on a returned unit, that's probably why.

Signoff:

```
prepared:   g.hoti     2026-02-14
reviewed:   a.karp     2026-02-15   (ext 4419, #ghoti-fw-bringup)
approved:   m.tuna     2026-02-17
```

Revision history:

```
rev 1   2025-08-22   g.hoti   first cut, copied from AKR-BU-SPI-010
rev 2   2025-11-04   a.karp   OTP-mirror offset 0xF80000 -> 0xFF0000
                              (per DVT-12 failure log, ticket GHT-FW-241)
rev 3   2026-01-19   m.tuna   added BCH8 note (verfied against
                              W25N01GV DS rev D)
```

TODO(g.hoti): confirm PEB0-lock semantics on B7 mask -- bootROM behaviour differs from B6, need silicon signoff before RTM. Might affect the OTP-mirror lock state at power-on.